

DIR-7
INDUSTRIAL ROBOT
ELECTRICAL CABINET



FUNCTIONAL FEATURES

- Adopt split design
G7 electrical cabinet is divided into power supply part and control part. The power supply part mainly focuses on heating devices so it adopts multi-group fan and air channel design to ensure good heat dissipation. The control part is isolated from the power supply part to prevent dust and oil pollution from entering and ensure the cleanliness of the control part. Besides, it can also avoid abnormal operation of the control part devices affected by dust and oil ollution.
- Equipped with safety emergency stop board independent of the control system, G7 electrical cabinet adopts imported forced disconnect relay to provide double circuit emergency stop to ensure the reliability of emergency stop.
- Automatic external power-on function
It can greatly ensure the safety of the operator through the methods of automatic external power-on function
- Energy-saving mode
It can effectively reduce standby energy consumption and avoid personal injury who is strayed in robots standby state.
- Adopt three-phase 380V power supply to direct power supply, which saves transformer and cost
- Adopt built-in three-phase filter to effectively isolate external interference and prevent internal interference output
- Double switching power supply design can avoid internal power interference
- It is convenient and simple to maintenance with split module design
- It can drive 50-220kg load with large output power.

CABINET TECHNICAL INDEX

Main power source	Phase/voltage	Main circuit power supply: three-phase 380VAC (-20%~ +10%), 50/60Hz Control loop power supply: single-phase AC220VAC (-10%~ +10%), 50/60Hz
	Input Power	Maximum Power:16KW
Cooling Mode	Air Cooling	
Insulation and Withstand Voltage	Grounding AC2600VAC@50Hz, voltage withstand 1Min (except control part of power supply)	
Institutions algorithm	Vertical multi-joint series, vertical multi-joint parallelogram, vertical multi-joint L-shaped wrist robot, etc	
System Mode	Teach, Reproduction, Remote	
Stopping Resistor	Internal	
Regenerative Resistor	External	
Instruction System	Motion, Logic, Process, Operation	
Soft PLC	Ladder diagram editing, 5000 steps, 10MS cycle	
Application	Handling, welding, spraying, palletizing, cutting, etc	
Teaching pendant	8-inch TFT-LCD, keyboard + touch screen, mode selection switch, safety switch, emergency stop button	
User memory	400M	
Communication Mode	TCP/IP ModbusTcp ModbusRtu CAN	
Controlled Axes	6+2 (The standard configuration is 6 axes, and the external axes need to be selected)	
Interference	Controller 22DI 22DO Drive Units 3DI 3DO	
	4-way 0~10V analog output, 12-bit accuracy(expandable COM)	
	Double encoder signal interface(position tracking)	
	Robot terminal: maintenance switch, external emergency stop	
	Ethernet CAN RS485 RS232	
Protective Function	2 USB interface	
	Overcurrent, overvoltage, undervoltage, overheat, overload, overspeed, excessive position deviation, abnormal communication, etc	
	Associated emergency stop can rapidly stop the robot when abnormal signal occurs	
	Installation Instruction	Indoor (avoid direct sunlight), no corrosive fog (avoid lampblack, flammable gas and dust)
	Altitude	Altitude: under 2000m
Environmental Specification	Ambient Temperature	-20℃~55℃ (If the ambient temperature exceeds 45℃, please keep the surrounding air to circulate
	Storage Temperature	-20℃~55℃ (Maximum temperature: 80℃ for 72 hours without condensation
	Humidity	Under 20~80 %RH No condensation
	Vibration	Random Vibration Frequency 20-500Hz, X Direction2.04m/s2, Y Direction7.4m/s2, Z Direction 10.4m/s2Sine Sweep Frequency 10-58.1Hz Acceleration under 10m/s2
	Connecting Cable	Under 20~93%RH No condensation
Dimension (MM)	630×939×472	
Weight (KG)	112KG	

CABINET DIMENSIONAL DRAWING

